From arithmetic dilogarithms to SICs in every dimension

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Abstract

Recent arithmetic results give a construction of symmetric informationally complete configurations in every finite complex dimension. This note explains the implication with explicit coordinates and phases. Twisted convolution produces a trace-one algebraic idempotent. The common-conjugation theorem of *Stark units for real quadratic fields and reciprocity laws* makes its nonzero displacement coefficients simultaneously have modulus one, so an algebraic conjugate is a Hermitian rank-one projector. Its displacement orbit consists of d^2 unit vectors with every distinct squared inner product equal to $1/(d+1)$. We include the even-dimensional representative convention and explicit vectors in dimensions two and three. The all-dimension consequence is already stated in the cited reciprocity preprint; the purpose here is to spell out its deduction from the arithmetic results.

1 The existence statement and its arithmetic sources

A symmetric informationally complete configuration, or SIC, in $\mathbb{C}^d$ is a family of d^2 unit vectors whose distinct squared inner products are all $1/(d+1)$. The lines they span are therefore equiangular. The existence of such a family for every integer $d \geq 2$ is the question recorded as AIM 030 [5]. The question imposes no group symmetry on the vectors.

Appleby, Flammia and Kopp [1] gave an arithmetic construction using special values of the Shintani–Faddeev modular cocycle. Their construction required both an algebraic conjugation predicted by Stark's conjectures and a twisted-convolution identity ensuring idempotency. Radchenko and Wheeler [2] proved algebraicity of the corresponding finite quantum dilogarithm values and a finite pentagon relation. Appleby, Flammia and Kopp [3] then proved the required convolution identity and supplied an explicit conversion between the two special-function conventions. These results produce algebraic rank-one idempotents, called ghost SIC projectors, but idempotency by itself does not make a matrix Hermitian.

The October preprint *Stark units for real quadratic fields and reciprocity laws* by Radchenko and Wheeler [4] proves the simultaneous conjugation needed for that last step. Its Proposition 7 states that every algebraic automorphism switching the two embeddings of the real quadratic field sends all the nonzero finite dilogarithm values to numbers of modulus one. The same source already states the SIC consequence in Section 9.2. We give a detailed deduction in the conventions of [1, 3, 2], with the coordinate conversion and all dimensions explicit.

Theorem 1 (Consequence of the cited arithmetic results)**.** *For every integer $d \geq 2$, there exist d^2 vectors $\psi_1, \ldots, \psi_{d^2}$ in $\mathbb{C}^d$ such that*

$$\psi_j^* \psi_j = 1, \qquad |\psi_j^* \psi_k|^2 = \frac{1}{d+1} \quad (j \neq k). \tag{1}$$

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Here star denotes conjugate transpose, so v^*w is conjugate linear in v. The construction for $d \geq 4$ uses a displacement orbit. This supplies witnesses for the unrestricted assertion (1). The argument concerns existence; it neither classifies all SICs nor provides a practical procedure for selecting the algebraic automorphism. Its arithmetic premises are the exact preprint versions cited below. No peer-review status is assumed.

The reason modulus one is sufficient is that the displacement coefficients already come in reciprocal pairs. After one common algebraic conjugation, the reciprocal of each coefficient becomes its complex conjugate. The idempotent is then Hermitian, and trace one makes its range a line. We first fix the displacement and special-function conventions, then derive the idempotent, and finally prove (1), including $d = 2, 3$. All algebraic numbers lie in a fixed algebraic closure $\overline{\mathbb{Q}} \subset \mathbb{C}$.

2 Displacements and finite arithmetic values

For $d \geq 2$, set

$$\omega_d = \exp(2\pi i/d), \qquad \xi_d = -\exp(\pi i/d), \qquad \bar{d} = \begin{cases} d, & d \text{ odd}, \\ 2d, & d \text{ even}. \end{cases}$$

In the standard basis $e_0, \ldots, e_{d-1}$ of $\mathbb{C}^d$, define $Xe_j = e_{j+1 \bmod d}$ and $Ze_j = \omega_d^j e_j$. For $p \in \mathbb{Z}^2$, put

$$D_p = \xi_d^{p_1 p_2} X^{p_1} Z^{p_2}, \qquad [p, q] = p_2 q_1 - p_1 q_2. \tag{2}$$

The identities $ZX = \omega_d XZ$ and $\xi_d^2 = \omega_d$ give

$$D_p D_q = \xi_d^{[p,q]} D_{p+q}, \qquad D_p^* = D_{-p}. \tag{3}$$

For any fixed representatives of $(\mathbb{Z}/d\mathbb{Z})^2$, the d^2 matrices D_p form a trace-orthogonal basis:

$$\operatorname{tr}(D_p) = 0 \quad (p \notin d\mathbb{Z}^2), \qquad \operatorname{tr}(D_p^* D_q) = d\,\delta_{p,q}. \tag{4}$$

A nontrivial shift has zero diagonal, and a nontrivial modulation has trace $\sum_{j=0}^{d-1} \omega_d^{p_2 j} = 0$, which proves these assertions. In even dimensions D_p itself can change sign on replacing p modulo d. Its projective class is well defined. Later sums use coefficients for which the entire summand is representative independent.

We next state the arithmetic result with the normalization needed here. Let K be a real quadratic field, $I \subset K$ a rank-two lattice, and $\varepsilon > 1$ a totally positive unit satisfying $\varepsilon I = I$. The prime denotes the other real embedding. Choose an oriented basis $I = \mathbb{Z}\beta_1 + \mathbb{Z}\beta_2$ with $\beta_1\beta_2' - \beta_1'\beta_2 > 0$ and β_2 totally positive. Put $t = \beta_1/\beta_2$, and let $A = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ satisfy $A(\beta_1, \beta_2)^T = \varepsilon(\beta_1, \beta_2)^T$. Then $A \cdot t = (at + b)/(ct + e) = t$ and $ct + e = \varepsilon$.

Write $\mathrm{e}(z) = \exp(2\pi i z)$ and $(z; q)_\infty = \prod_{n \geq 0}(1 - zq^n)$ for $|q| < 1$. The modular quantum dilogarithm is initially the quotient

$$\Phi_{A,m,n}(z; t) = \frac{(\mathrm{e}(mt + z); \mathrm{e}(t))_\infty}{(\mathrm{e}(nA \cdot t + z/(ct + e)); \mathrm{e}(A \cdot t))_\infty} \qquad (\operatorname{Im} t > 0). \tag{5}$$

At the real fixed point it means the meromorphic continuation described in [2, Section 2.2], rather than direct substitution into either infinite product. Let μ_A be the Dedekind eta multiplier, defined in the upper half-plane by

$$\eta(A \cdot t) = \mu_A \sqrt{ct + e}\, \eta(t), \qquad \eta(t) = \mathrm{e}(t/24) \prod_{n \geq 1} (1 - \mathrm{e}(nt)),$$

with the principal square root. This multiplier is a root of unity.

For a nonzero class in the finite group $\mathcal{G}_{I,\varepsilon} = (\varepsilon-1)^{-1}I/I$, choose a representative x and integers m, n such that $(\varepsilon^{-1}-1)x/\beta_2 = mt+n$. Define

$$E_{I,\varepsilon}(x) = \mu_A \Phi_{A,m,0}(x/\beta_2; t), \qquad E_{I,\varepsilon}(0) = \sqrt{\varepsilon} > 0. \tag{6}$$

The algebraicity, reflection and continuation results of [2, Theorems 1–2], expressed in these lattice coordinates in [4, Section 3 and Appendix B], make the values nonzero algebraic numbers and independent of the permitted choices.

Proposition 2 (Common conjugation, [4, Proposition 7])**.** *For every K, I, ε as above and every automorphism τ of $\overline{\mathbb{Q}}$ restricting to the nontrivial automorphism of K, one has*

$$|\tau(E_{I,\varepsilon}(x))| = 1 \quad \textit{for all } x \in \mathcal{G}_{I,\varepsilon} \setminus \{0\}. \tag{7}$$

The quantifier over τ is shared by all x in this statement. There is no restriction to a maximal-order lattice or to a fundamental unit. This allows the lattice and the third power of a unit used below.

3 A trace-one arithmetic idempotent

Fix $d \geq 4$, and set

$$\Delta = (d+1)(d-3), \quad \rho = \frac{d-1+\sqrt{\Delta}}{2}, \quad K = \mathbb{Q}(\sqrt{\Delta}), \quad I = \mathbb{Z}\rho + \mathbb{Z}, \quad \varepsilon = \rho^3. \tag{8}$$

We have $(d-2)^2 < \Delta < (d-1)^2$, since the respective differences are $2d-7 > 0$ and 4. Thus Δ is positive and nonsquare. Moreover, $\rho^2 - (d-1)\rho + 1 = 0$ and $\rho' = \rho^{-1} > 0$. Hence ρ is a totally positive integral unit preserving I, and $(\rho, 1)$ is oriented. Multiplication by ρ and by ε is represented respectively by

$$L = \begin{pmatrix} d-1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A = L^3 = \begin{pmatrix} d^3-3d^2+d+1 & -d(d-2) \\ d(d-2) & 1-d \end{pmatrix}. \tag{9}$$

In particular,

$$\frac{A - I_2}{d} = \begin{pmatrix} d^2-3d+1 & 2-d \\ d-2 & -1 \end{pmatrix}, \qquad N = \operatorname{tr}(A) - 2 = d^2(d-3) > 0. \tag{10}$$

The fixed point of A is ρ and its lower-left entry is positive.

Here is the precise rank-one instance of the dimension correspondence in [1, Definitions 1.28–1.30 and Theorem 4.50]. The integral form

$$Q_d(x, y) = x^2 - (d-1)xy + y^2$$

has discriminant Δ. The tuple $(d, 1, Q_d)$ is admissible in [3, Definition 1.1]: $1 < (d-1)/2$, $(d^2-1)/(d-1) = d+1$ is an integer, and $\Delta/\operatorname{disc}(Q_d) = 1$. In the notation of [1], ρ is the jth power of the fundamental totally positive unit, $d = d_{j,1}$, and the conductor of the form is $f = f_j$. Indeed, if Δ_0 is the field discriminant, this conductor is $(\rho - \rho')/\sqrt{\Delta_0}$. The source matrix $L_{z,t}$ is exactly L, and its rank-one stabilizer $L_{z,t}^3$ is A.

For $p \in \mathbb{Z}^2$ put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad u = S^{-1}(A - I_2)p/d, \qquad x_p = \frac{\rho p_2 - p_1}{d} \pmod{I}. \tag{11}$$

The vector u is integral by (10), with entries

$$u_1 = (d-2)p_1 - p_2, \qquad u_2 = -(d^2 - 3d + 1)p_1 + (d-2)p_2.$$

Using $A + A^{-1} = (N+2)I_2$ and $\rho^2 = (d-1)\rho - 1$ gives

$$\frac{(I_2 - A^{-1})Su}{N} = \frac{p}{d}, \qquad \frac{u_1\rho + u_2}{\varepsilon^{-1} - 1} = \frac{\rho p_2 - p_1}{d}. \tag{12}$$

Since $A \equiv I_2 \pmod{d}$, the class x_p belongs to $\mathcal{G}_{I,\varepsilon}$; it is zero exactly when $p \in d\mathbb{Z}^2$. The nonzero branch of the special-function dictionary applies at precisely these indices. To check its exclusion, put $M = (I_2 - A^{-1})S/N$. Then $Mu = p/d$ and

$$M(A^T - I_2)\mathbb{Z}^2 = -A^{-1}S\mathbb{Z}^2 = \mathbb{Z}^2,$$

so $u \in (A^T - I_2)\mathbb{Z}^2$ if and only if $p \in d\mathbb{Z}^2$.

For $p \notin d\mathbb{Z}^2$, define

$$s_d(p) = d + (1+d)(1+p_1)(1+p_2), \qquad \nu_p = \frac{(-1)^{s_d(p)}\xi_d^{-Q_d(p)}}{E_{I,\varepsilon}(x_p)}. \tag{13}$$

These are the normalized ghost coefficients of [1]. We spell out the normalization because an uncancelled multiplier would change the convolution. Write $\mathcal{S}_A^r(\rho)$ for the Shintani–Faddeev modular cocycle at rational column index r, in the convention of [1]. On the nonzero branch, [3, Proposition 2.1] gives

$$F_A(u^T) = \mu_A\, \mathcal{S}_A^{Mu}(\rho)^{-1} = E_{I,\varepsilon}(x_p). \tag{14}$$

The source phase is $\phi_p = (-1)^{s_d(p)}\mu_A^{-1}\xi_d^{-Q_d(p)}$, because $f = f_j$ and the rank is one. Thus $\phi_p \mathcal{S}_A^{p/d}(\rho) = \nu_p$. The zero coefficient is separately defined by

$$\nu_0 = -\varepsilon^{-1/2} \qquad [1, \text{Equation (5.65)}]. \tag{15}$$

It is not obtained by inserting the zero class into (13). The unconditional reciprocal identity is

$$\nu_p \nu_{-p} = 1 \quad (p \notin d\mathbb{Z}^2) \qquad [1, \text{Theorem 5.8}]. \tag{16}$$

Lemma 3. *Let $G = \operatorname{diag}(-1, 1)$. Then*

$$B = \frac{I_d}{d} + \frac{1}{d\sqrt{d+1}} \sum_{p \neq 0 \bmod d} \nu_{Gp} D_p \tag{17}$$

is an algebraic matrix satisfying $B^2 = B$ and $\operatorname{tr}(B) = 1$. The entire summands in this expression are independent of the chosen integer representatives.

Proof. The proved convolution theorem [3, Theorem 1.2] permits a shift λ with $\lambda r_{j,m} \equiv -r_{j,m-1} \pmod d$, as specified in its proof on pages 7–8. At $m = 1$ we have $r_{j,0} = 0$ and $r_{j,1} = 1$, so $\lambda = 0$ is available. With the inverse cocycle identity [1, Lemma 2.13], this gives

$$\sum_q \omega_d^{[p,Lq]} \frac{\mathcal{S}_A^{q/d}(\rho)}{\mathcal{S}_A^{(q-p)/d}(\rho)} = 0 \qquad (p \neq 0 \bmod d). \tag{18}$$

The representatives in this sum include 0 and p. We use only this proved shift; no assertion about the entire set of conjectural shifts is needed.

The twist condition of [1, Lemma 1.42] is $\det(G)(2\lambda + d - 1 + d) \equiv 1 \pmod{\bar{d}}$. For our choices its left side is $1 - 2d$, so it holds in both parities. The same lemma gives representative independence of $\nu_{Gp} D_p$.

For the phases ϕ_q above, direct calculation yields

$$\xi_d^{-[p,q]} \frac{\phi_q}{\phi_{q-p}} = (-1)^{1+s_d(p)} \xi_d^{Q_d(p)} \omega_d^{[p,Lq]}. \tag{19}$$

Indeed,

$$\begin{aligned} s_d(q) + s_d(q-p) &\equiv 1 + s_d(p) + (1+d)[p,q] \pmod 2, \\ Q_d(q-p) - Q_d(q) &= Q_d(p) - (d-1)[p,q] + 2[p,Lq]. \end{aligned}$$

Together with $\xi_d^d = (-1)^{d+1}$ these identities prove (19). Multiplying (18) by the common nonzero phase therefore gives

$$\sum_q \xi_d^{-[p,q]} \frac{\nu_q}{\nu_{q-p}} = 0. \tag{20}$$

The zero values in this normalization are exactly (15). By (16), the $q = 0, p$ terms are $\nu_p(\nu_0 + \nu_0^{-1})$. Since $\sqrt{\rho} + 1/\sqrt{\rho} = \sqrt{d+1}$, we obtain

$$\nu_0 + \nu_0^{-1} = -(\rho^{3/2} + \rho^{-3/2}) = -(d-2)\sqrt{d+1}.$$

Consequently

$$\sum_{q \neq 0,p} \xi_d^{-[p,q]} \nu_q \nu_{p-q} = (d-2)\sqrt{d+1}\, \nu_p. \tag{21}$$

Put $F = \sum_{p \neq 0} \nu_p D_{G^{-1}p}$. Since $\det(G^{-1}) = -1$, (3) gives the phase $\xi_d^{-[p,q]}$ in the coefficient of $D_{G^{-1}p}$ in F^2. Formula (21) computes that coefficient. The identity coefficient is $d^2 - 1$, by (16). Thus

$$F^2 = (d^2 - 1) I_d + (d-2)\sqrt{d+1}\, F. \tag{22}$$

Representative independence justifies the changes of index also when d is even. Substitution in $B = I_d/d + F/(d\sqrt{d+1})$ proves $B^2 = B$. Equation (4) gives $\operatorname{tr}(B) = 1$. All its entries are algebraic, by the arithmetic algebraicity theorem and (13). No Hermiticity assertion has been used. □

4 Hermitian conjugation and the vector family

The nontrivial automorphism of K extends to an automorphism τ of $\overline{\mathbb{Q}}$. For completeness, order partial field embeddings into $\overline{\mathbb{Q}}$ extending that automorphism by extension of their domains. Unions of chains are embeddings, so Zorn's lemma gives a maximal one. If its domain omits an algebraic

number, a root of the transformed minimal polynomial extends the embedding, a contradiction. Its domain is therefore $\overline{\mathbb{Q}}$. Its image is algebraically closed: pull back a polynomial with coefficients in the image, take a root, and map that root. An algebraically closed subfield of $\overline{\mathbb{Q}}$ containing $\mathbb{Q}$ contains every root of every rational polynomial, so the image is all of $\overline{\mathbb{Q}}$. This proves the required automorphism extension. It may be selected separately for each d.

Apply Proposition 2 to this one τ and all x_{Gp}. Roots of unity map to roots of unity, so

$$c_p = \tau(\nu_{Gp}), \qquad |c_p| = 1, \qquad c_p c_{-p} = 1, \qquad c_{-p} = \overline{c_p}. \tag{23}$$

We must also conjugate the displacement matrices and the normalization. There are k coprime to $\bar{d}$ and $h \in \{1, -1\}$ with

$$\tau(\xi_d) = \xi_d^k, \qquad \tau(\sqrt{d+1}) = h\sqrt{d+1}.$$

Since X has rational entries and $\tau(Z) = Z^k$, entrywise conjugation gives

$$U_p = \tau(D_p) = D_{(p_1, kp_2)}, \qquad P = \tau(B) = \frac{I_d}{d} + \frac{h}{d\sqrt{d+1}} \sum_{p \neq 0} c_p U_p. \tag{24}$$

In particular $U_{-p} = U_p^*$, and $p \mapsto (p_1, kp_2)$ permutes the classes modulo d. The full summands $c_p U_p$ remain representative independent. Reindexing their adjoints by $p \mapsto -p$ and using (23) proves $P^* = P$. This argument includes self-inverse classes in even dimensions. It does not require τ to commute with complex conjugation on arbitrary algebraic numbers.

Applying a field automorphism to the algebraic equations in Lemma 3 gives $P^2 = P$ and $\operatorname{tr}(P) = 1$. Every idempotent has the direct sum decomposition $\mathbb{C}^d = \operatorname{ran} P \oplus \ker P$, and is diagonal with ones on its range and zeros on its kernel in an adapted basis. Therefore $\operatorname{rank} P = \operatorname{tr}(P) = 1$. Hermiticity makes the kernel orthogonal to the range. Choose a unit vector ψ spanning that range. Then

$$P = \psi\psi^*, \qquad v^* P v = \|Pv\|^2 \geq 0 \quad (v \in \mathbb{C}^d). \tag{25}$$

For every nonzero class q modulo d, trace orthogonality and the permutation in (24) imply

$$|\operatorname{tr}(P D_q^*)| = \frac{1}{\sqrt{d+1}}, \qquad |\psi^* D_q \psi|^2 = \frac{1}{d+1}. \tag{26}$$

Changing an integer representative only multiplies the relevant displacement by a phase of modulus one. The possible sign h likewise disappears from the absolute value. For one representative of each $q \in (\mathbb{Z}/d\mathbb{Z})^2$ define $\psi_q = D_q \psi$. All these vectors have unit norm. If $q \neq r$, then $D_q^* D_r$ is a unit phase times D_{r-q}, and hence

$$|\psi_q^* \psi_r|^2 = |\psi^* D_{r-q} \psi|^2 = \frac{1}{d+1}.$$

Since this number is less than one, no two vectors are parallel. We have therefore obtained exactly d^2 vectors satisfying (1) for every $d \geq 4$.

The two remaining dimensions have elementary witnesses. For $d = 2$, set $\zeta = \exp(2\pi i/3)$ and take

$$v_0 = (1, 0), \qquad v_{j+1} = (1/\sqrt{3}, \sqrt{2/3}\,\zeta^j), \quad j = 0, 1, 2.$$

Their norms are one. The squared overlaps with v_0 equal $1/3$, and for $j \neq k$ the other squared overlaps equal $|1 + 2\zeta^{k-j}|^2/9 = 1/3$, since $\operatorname{Re}\zeta = \operatorname{Re}\zeta^2 = -1/2$.

For $d = 3$, take $v = (0, 1, -1)/\sqrt{2}$ and the nine vectors $X^p Z^q v$, with $p, q = 0, 1, 2$ and the dimension-three generators defined in Section 2. For $q = 1, 2$,

$$v^* Z^q v = (\omega_3^q + \omega_3^{2q})/2 = -1/2.$$

For $p = 1$ the overlap $v^* X^p Z^q v$ is $-\omega_3^q/2$, and for $p = 2$ it is $-\omega_3^{2q}/2$. Thus every nontrivial shift or modulation has overlap of modulus $1/2$. Weyl multiplication reduces every distinct pair in the orbit to one of these overlaps, so its squared inner product is $1/4$. Together with the $d \geq 4$ construction, these two cases prove Theorem 1 for every integer $d \geq 2$.

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